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LAB 5 – STATIC CODE ANALYSIS

Food Delivery Order Queue Manager
Screenshot Evidence Report

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Summary of Results

Tool	Before	After
Pylint	6.27/10	10.00/10
Bandit	3 findings	0 findings
Flake8	12 violations	0 violations

Screenshot 1 – Original Pylint Report

Baseline Pylint analysis identified multiple code-quality and maintainability issues. The original score was 6.27/10.

```
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery> Get-Content pylint_report_before.txt
************* Module delivery_system
delivery_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
delivery_system.py:7:0: C0103: Constant name "next_id" doesn't conform to UPPER_CASE naming style (invalid-name)
delivery_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:11:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
delivery_system.py:12:4: W0603: Using the global statement (global-statement)
delivery_system.py:21:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:24:4: W0702: No exception type(s) specified (bare-except)
delivery_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:28:4: W0603: Using the global statement (global-statement)
delivery_system.py:32:11: W0718: Catching too general exception Exception (broad-exception-caught)
delivery_system.py:30:12: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
delivery_system.py:30:12: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:42:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:43:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:47:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:48:4: W0603: Using the global statement (global-statement)
delivery_system.py:49:25: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
delivery_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
delivery_system.py:51:0: C0103: Function name "printQueue" doesn't conform to snake_case naming style (invalid-name)
delivery_system.py:52:10: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
delivery_system.py:56:0: C0116: Missing function or method docstring (missing-function-docstring)

-----------------------------------
Your code has been rated at 6.27/10
```

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Screenshot 2 – Original Bandit Report

Baseline Bandit analysis identified three findings: B403 (pickle import), B110 (try/except/pass), and B301 (pickle deserialization).

```
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery> Get-Content bandit_report_before.txt
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.13.15
Run started:2026-10-31 03:33:46.114766+00:00

Test results:
>> Issue: [B403:blacklist] Consider possible security implications associated with pickle module.
   Severity: Low   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_imports.html#b403-import-pickle
   Location: .\delivery_system.py:2:0
1       import json
2       import pickle
3       from datetime import datetime

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\delivery_system.py:24:4
23              orders[order_id]["status"] = status
24          except:
25              pass
26

--------------------------------------------------
>> Issue: [B301:blacklist] Pickle and modules that wrap it can be unsafe when used to deserialize untrusted data, possible security issue.
   Severity: Medium   Confidence: High
   CWE: CWE-502 (https://cwe.mitre.org/data/definitions/502.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b301-pickle
   Location: .\delivery_system.py:49:13
48          global orders
49          orders = pickle.load(open(file, "rb"))
50
```

Screenshot 3 – Original Flake8 Report

Baseline Flake8 analysis reported 12 style and formatting violations.

```
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery> Get-Content flake8_report_before.txt
delivery_system.py:11:1: E302 expected 2 blank lines, found 1
delivery_system.py:16:80: E501 line too long (96 > 79 characters)
delivery_system.py:21:1: E302 expected 2 blank lines, found 1
delivery_system.py:24:5: E722 do not use bare 'except'
delivery_system.py:27:1: E302 expected 2 blank lines, found 1
delivery_system.py:35:1: E302 expected 2 blank lines, found 1
delivery_system.py:42:1: E302 expected 2 blank lines, found 1
delivery_system.py:47:1: E302 expected 2 blank lines, found 1
delivery_system.py:51:1: E302 expected 2 blank lines, found 1
delivery_system.py:54:80: E501 line too long (80 > 79 characters)
delivery_system.py:56:1: E302 expected 2 blank lines, found 1
delivery_system.py:70:1: E305 expected 2 blank lines after class or function definition, found 1
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery>
```

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Screenshot 4 – Final Program Execution

The refactored food delivery order manager executed successfully and displayed delivered-order revenue of 440.

```
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery> python delivery_system.py
Revenue today: 440
Order queue at 2026-10-31 09:30:13.403462
1 Asha delivered 290
2 Ravi out_for_delivery 240
3 Meera delivered 150
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery>
```

Screenshot 5 – Final Pylint Report

The updated source code achieved a Pylint score of 10.00/10.

```
(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery> Get-Content pylint_report_after.txt

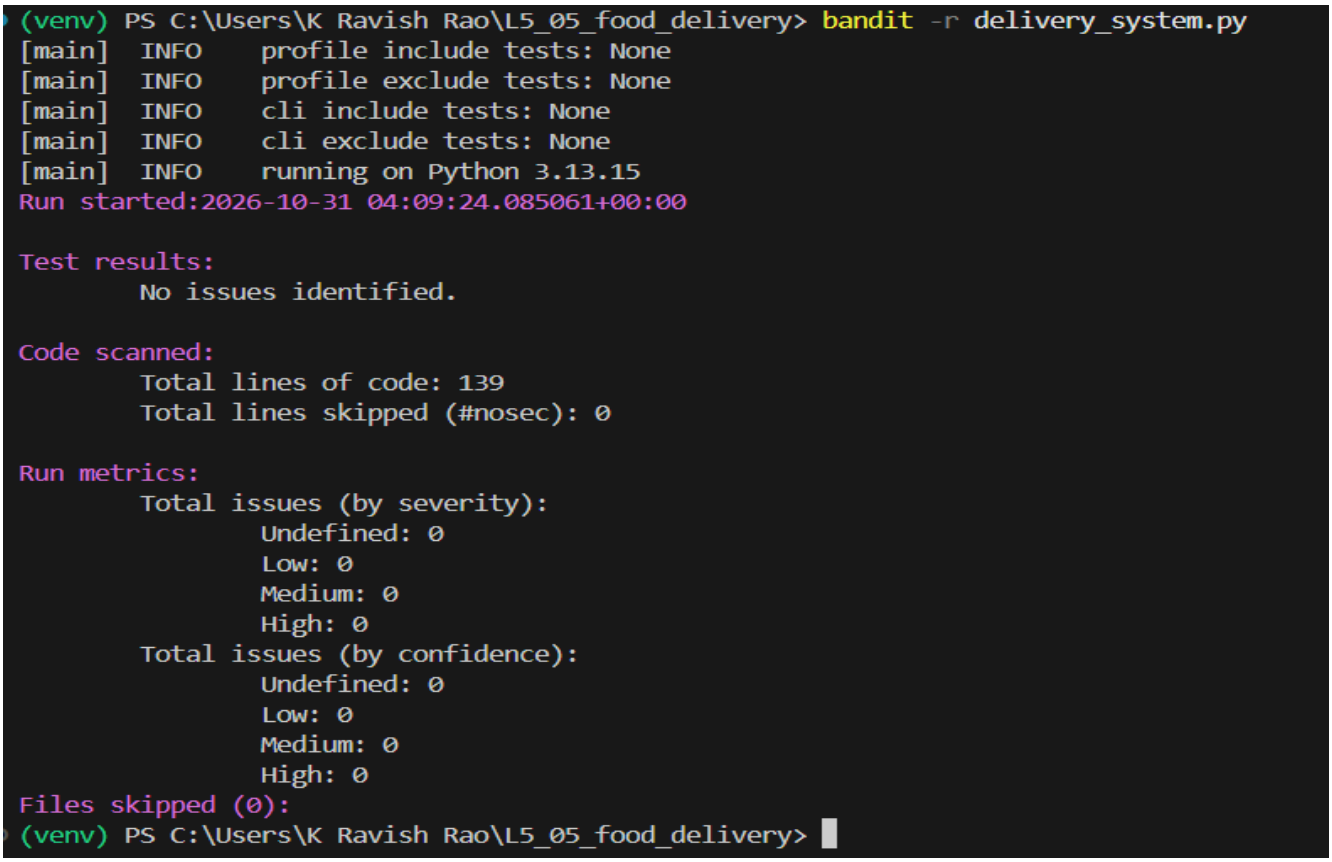
-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

(venv) PS C:\Users\K Ravish Rao\L5_05_food_delivery>
```

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Screenshot 6 – Final Bandit Report

The final Bandit scan reported no issues identified and zero findings across severity levels.



Screenshot 7 – Final Flake8 Report

The final Flake8 report was empty, indicating no reported style violations.

